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Set 18: Product Inventory Management

Aim

To analyse product inventory in R by creating a bar chart of product quantity, a stacked bar chart by category, an inventory table, and a combined dashboard.

Questions (combined into one experiment)

1. Bar chart of quantity of each product (axes labelled, titled)
2. Stacked bar chart of quantity by product category
3. Table of inventory data for each product
4. Dashboard combining the bar chart, stacked bar chart and table

Procedure

1. Open RStudio and install/load the packages ggplot2 and gridExtra.
2. Create the dataset as a data frame.
3. Draw the charts for Questions 1 and 2 using ggplot2 with titles and axis labels.
4. Display the dataset as a table using print() and tableGrob().
5. Combine the two charts and the table into a dashboard using grid.arrange().
6. Run the program and observe the output.

Code

```
library(ggplot2)
library(gridExtra)

# Dataset (Category column added for the stacked bar chart)
df <- data.frame(ProductID = 1:3, ProductName = c("Product A", "Product B", "Product C"),
                 Quantity = c(250, 175, 300), Price = c(20, 15, 18),
                 Category = c("Category 1", "Category 1", "Category 2"))

# Q1. Bar chart of quantity per product
p1 <- ggplot(df, aes(x = ProductName, y = Quantity)) + geom_col(fill = "darkgreen") +
  labs(title = "Quantity of Each Product", x = "Product", y = "Quantity Available")
print(p1)

# Q2. Stacked bar chart by category
p2 <- ggplot(df, aes(x = Category, y = Quantity, fill = ProductName)) + geom_col() +
  labs(title = "Quantity by Product Category", x = "Category", y = "Quantity", fill = "Product")
print(p2)

# Q3. Table of inventory data
print(df)
tb <- tableGrob(df, rows = NULL)
```

```
# Q4. Dashboard (bar chart + stacked bar chart + table)
grid.arrange(p1, p2, tb, ncol = 2, top = "Product Inventory Dashboard")
```

Output

Table (Question 3):

	ProductID	ProductName	Quantity	Price	Category
1	1	Product A	250	20	Category 1
2	2	Product B	175	15	Category 1
3	3	Product C	300	18	Category 2

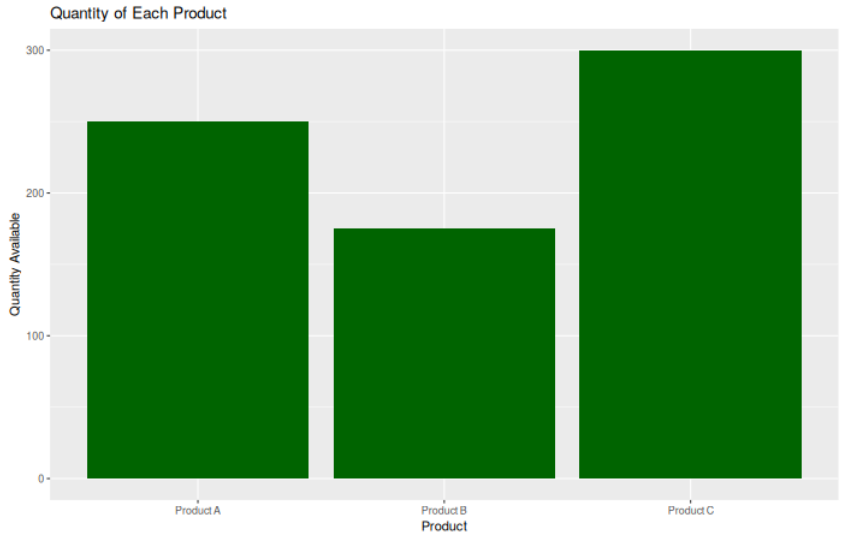


Figure 1: Question 1

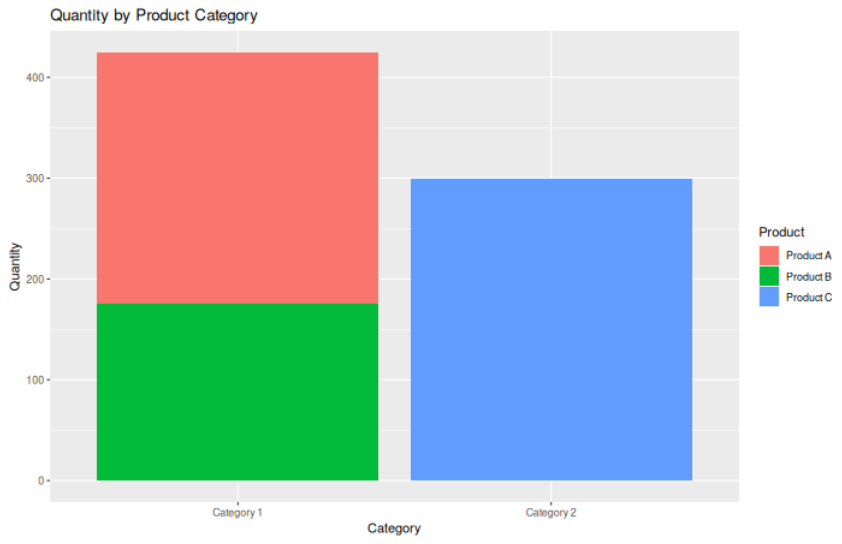


Figure 2: Question 2



ProductID	ProductName	Quantity	Price	Category
1	Product A	250	20	Category 1
2	Product B	175	15	Category 1
3	Product C	300	18	Category 2

Figure 3: Dashboard (Question 4)

Result

The bar chart, stacked bar chart, inventory table and combined dashboard for product inventory were created successfully using R.